

HPC Worksop: From basics to performance

Presented by

Raweeroj Thongdee

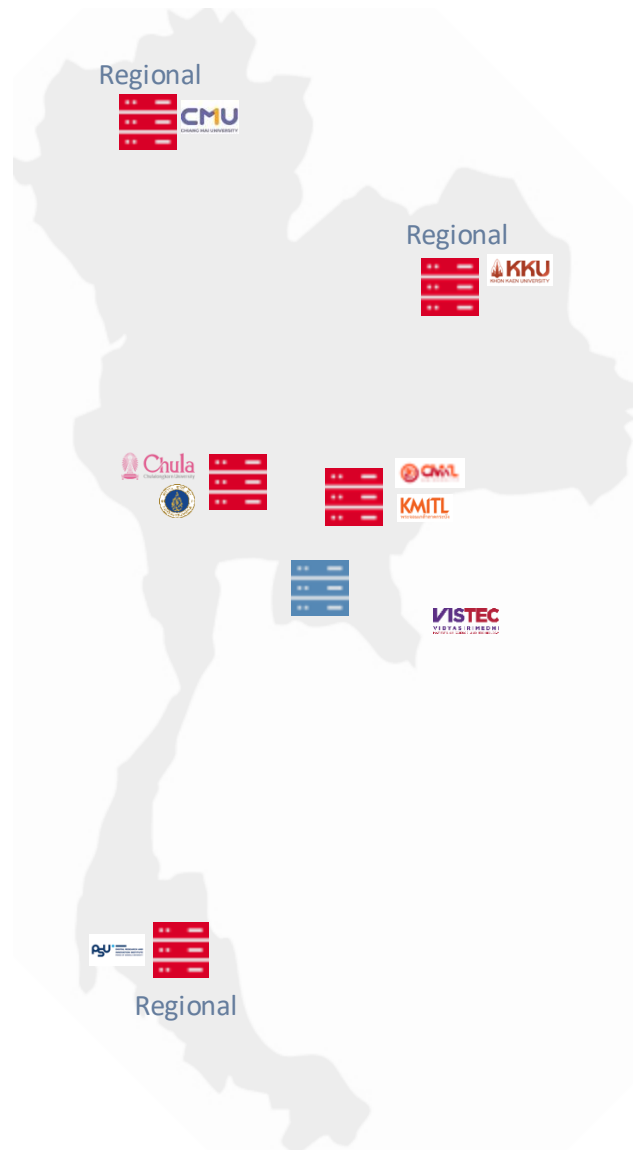
HPC System Engineer | IT Infrastructure

CMKL University



16th May 2025

HPC ในประเทศไทย ปัจจุบัน



AI Infrastructure
(AIEI)



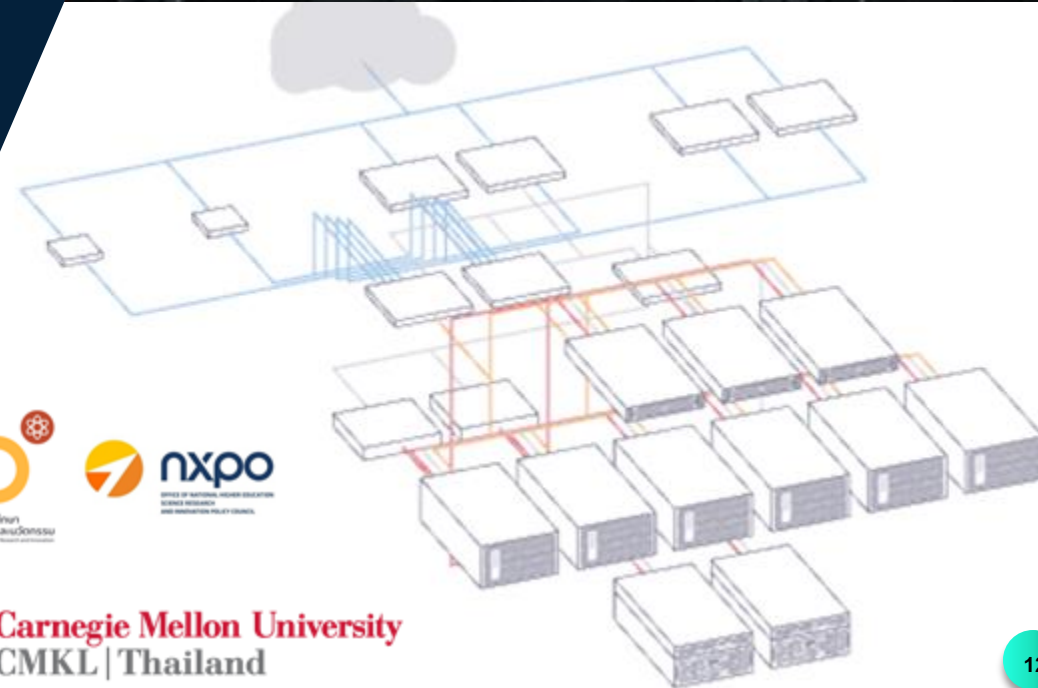
NSTDA Supercomputer Center
ThaiSC

โครงสร้างพื้นฐานสำหรับปัญญาประดิษฐ์ AI Infrastructure: APEX ที่ใช้งานอยู่ในปัจจุบัน

Apex: AI Research Platform & Data Exchange

R&D Support from PMU-C, Ministry of Higher Education, Science, Research and Innovation

- **Apex** - AI Supercomputer
 - First node in Thailand university AI research infrastructure
 - Partnership with NVIDIA and DDN for Technology Provider
- System overview - 6x DGX A100
 - 30 petaFLOPS AI (FP16) / 936 teraFLOPS (FP64)
 - 1920 cores / 48 A100 GPUs
 - 6 TB system memory + 1.92 TB GPU memory
 - 3 PB storage (434 TB SSD / 1.9 PB HDD usable storage)
 - 200 Gbps InfiniBand Interconnect
 - 100 GbE Open Ethernet



UWU.

มหาวิทยาลัยอุบลราชธานี
Ubon Ratchathani University

Carnegie Mellon University
CMKL | Thailand

ความพร้อมในการดำเนินงาน

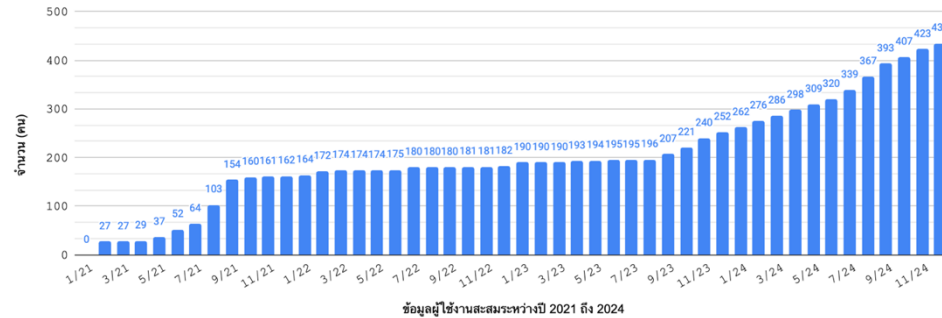
โครงสร้างพื้นฐานสำหรับปัญญาประดิษฐ์ AI Infrastructure: APEX ที่ใช้งานอยู่ในปัจจุบัน

ผู้ใช้งานในระบบ

435



จำนวนผู้ใช้งาน (Users)



Peak Utilization

90%



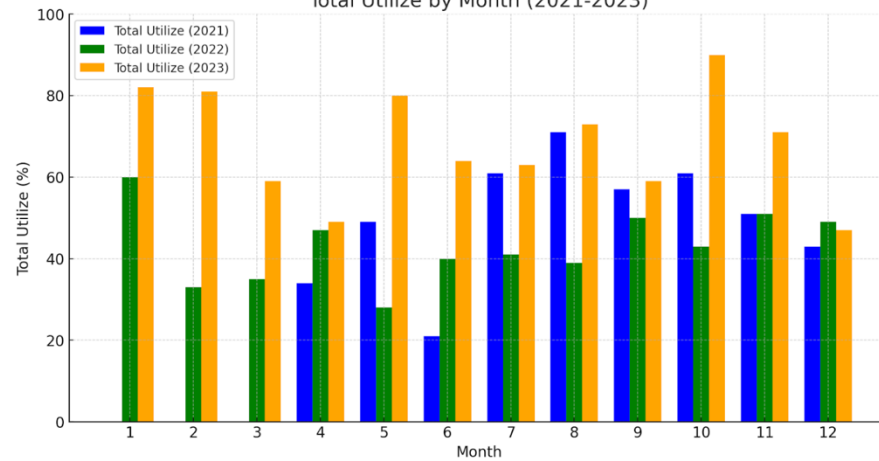
GPU Time

25M



Unit x Duration (Hour.)

Total Utilize by Month (2021-2023)



บริการให้กับหน่วยงานมากกว่า 50+ หน่วยงาน



ในส่วนของ APEX

ด้านสถานที่ - พื้นที่

CMKL Datacenter



Power source	: 1	: KMITL-Substation 115/24 kV
Maximum Power (IT)	: 125 kW	
UPS	: 120 kW	
Cooling	: Air-cooling	
	: Hot-Aside containment	
	: 140 kW (35kW x 4)	
	: T. 21-24C / H. 48-52%	
Tier	: Tier III Uptime	
Location	: KLLC Building B Floor 7, KMITL	



ระบบจ่ายพลังงาน



ระบบปรับอากาศ

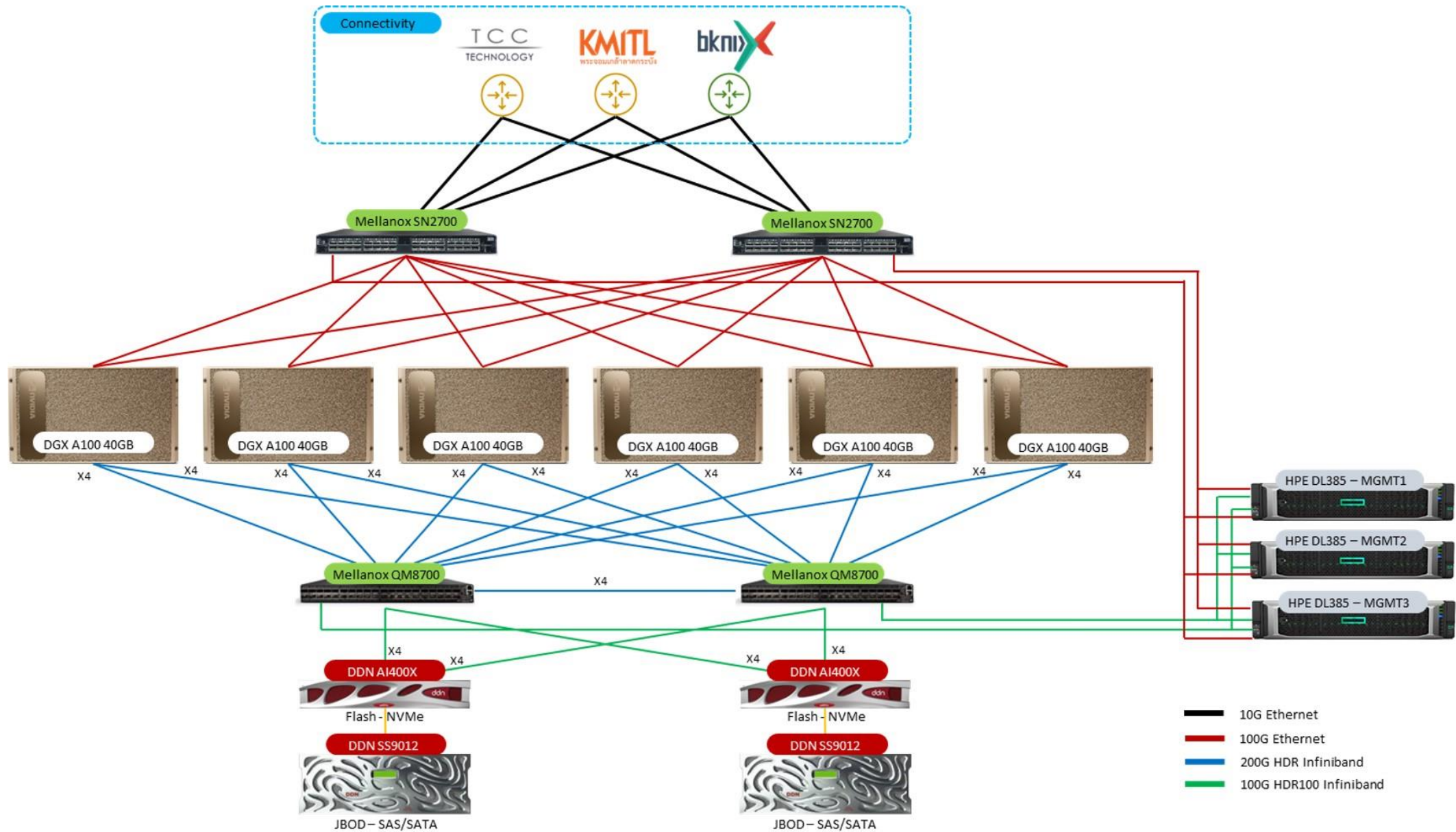
Datacenter

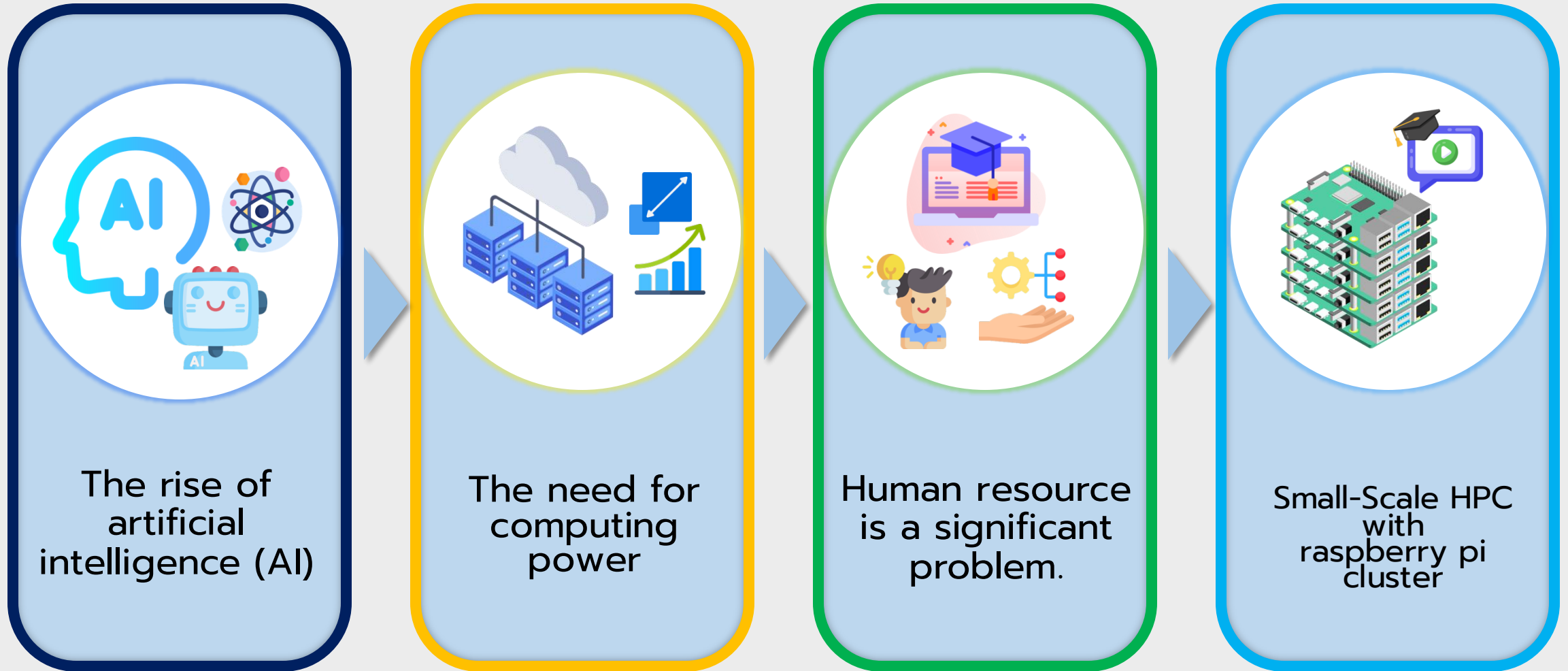
ตำแหน่งที่ - พื้นที่



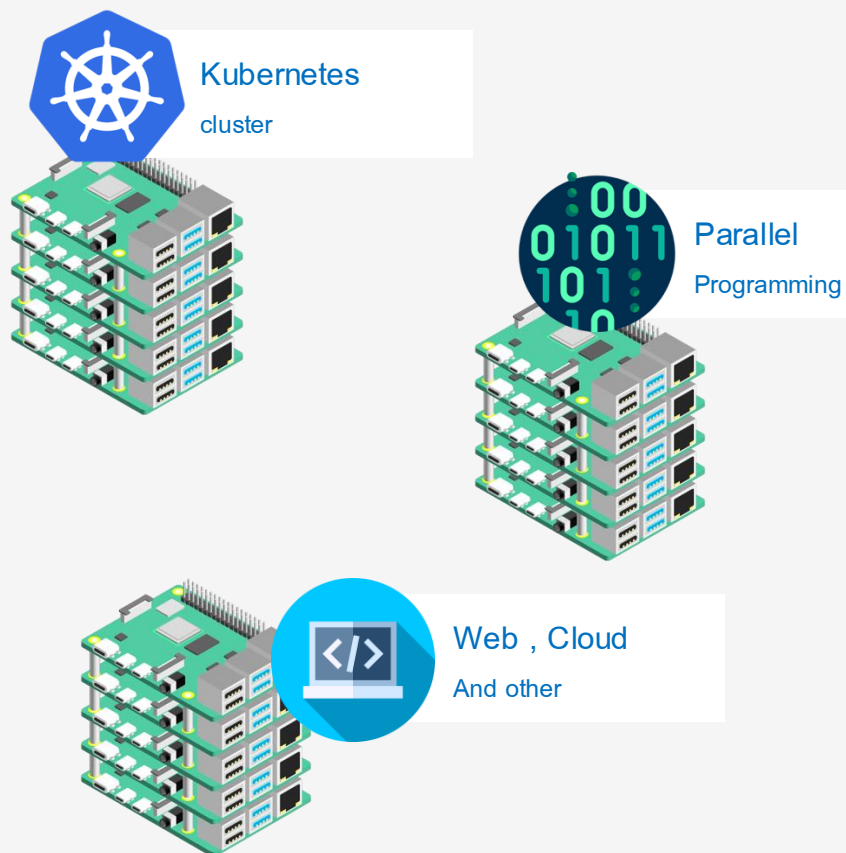
พื้นที่จัดเก็บอุปกรณ์และครุภัณฑ์

Datacenter



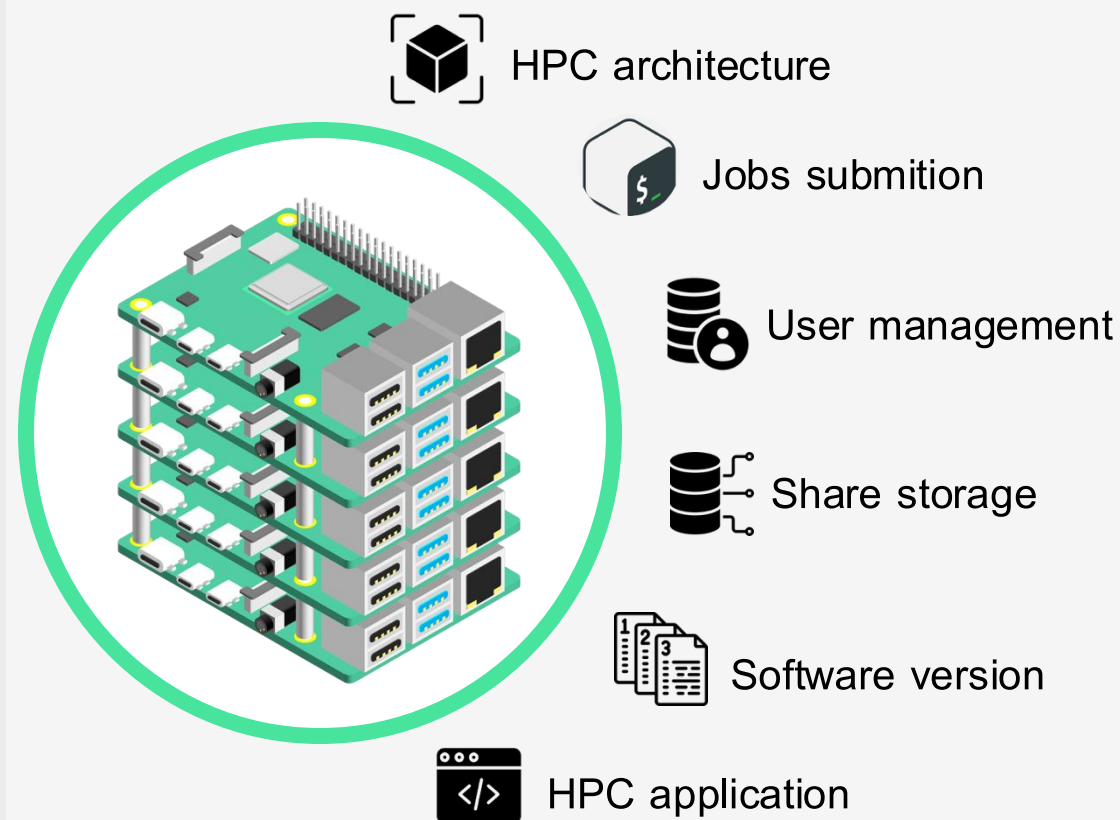


Previous

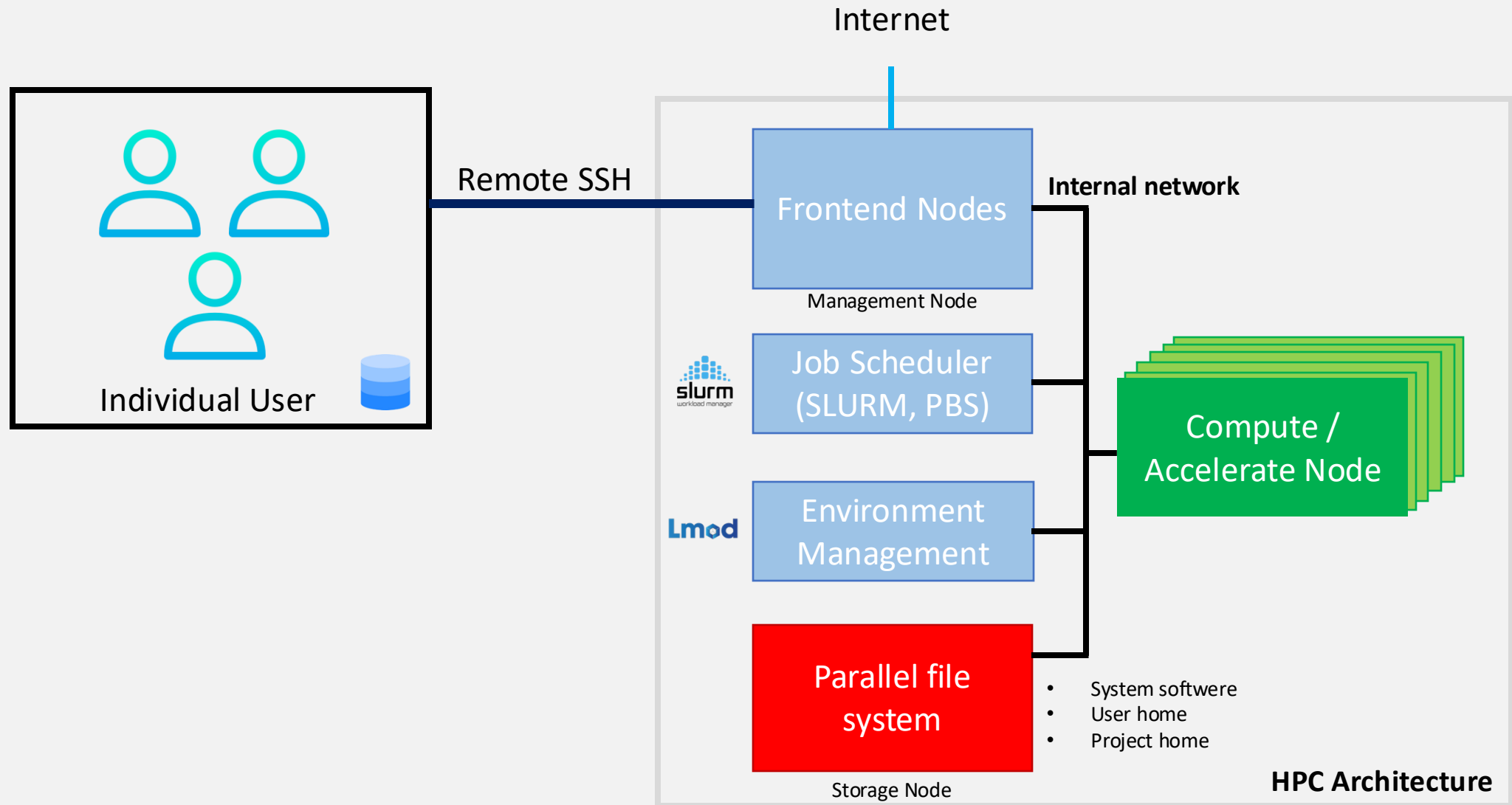


The background research topic of the Raspberry Pi cluster

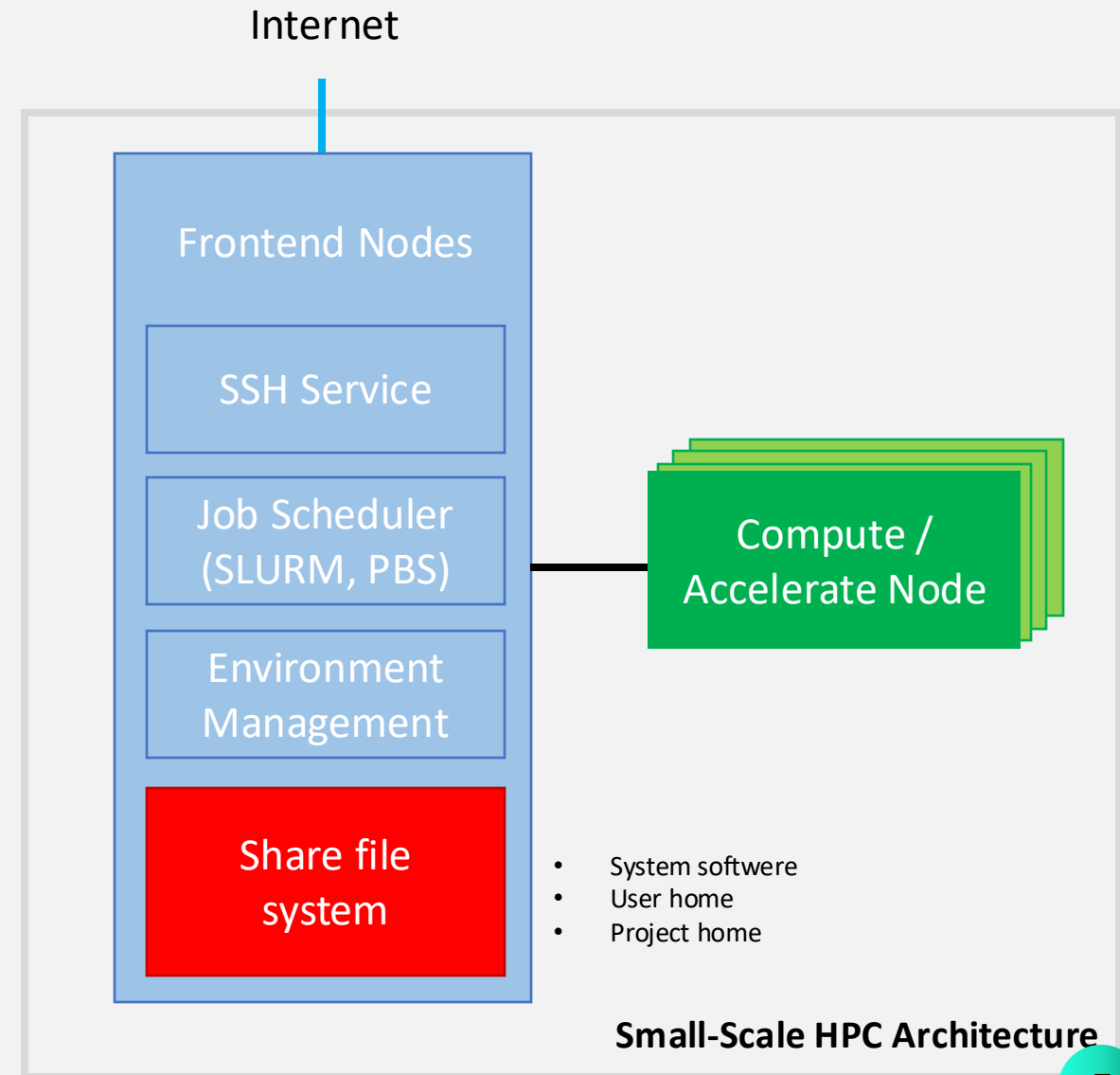
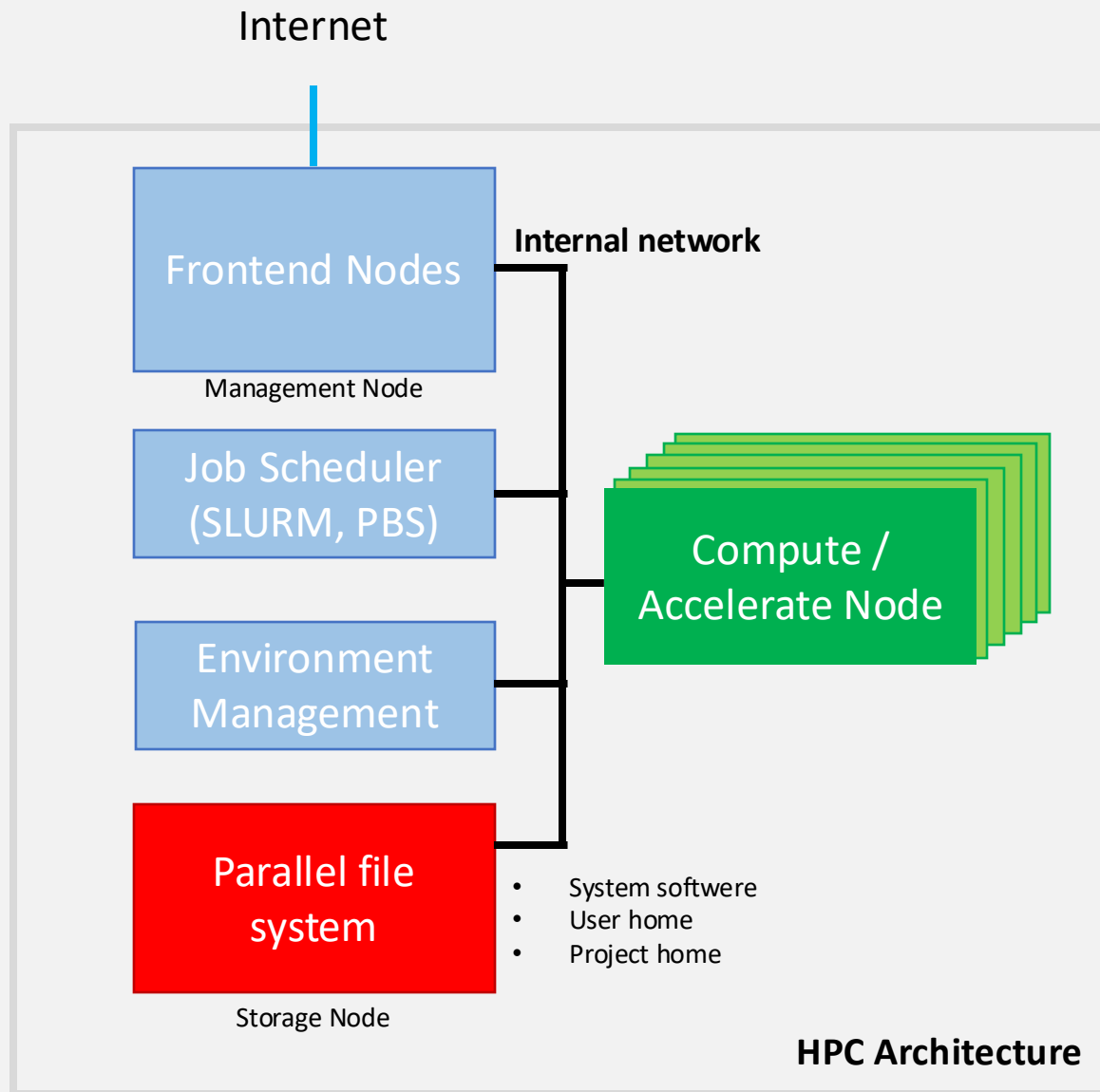
Present



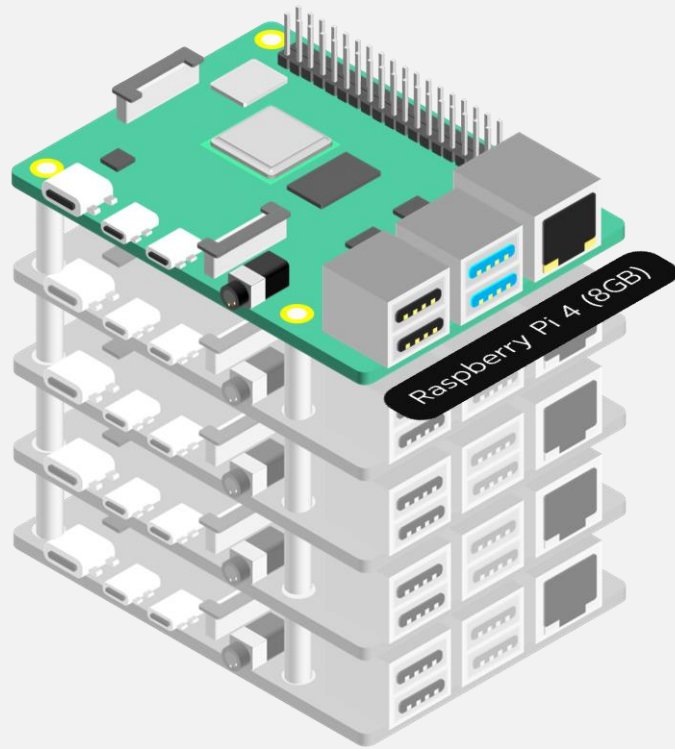
Small-Scale HPC with Raspberry Pi cluster



Design and Method – Overview about Small-Scale HPC systems



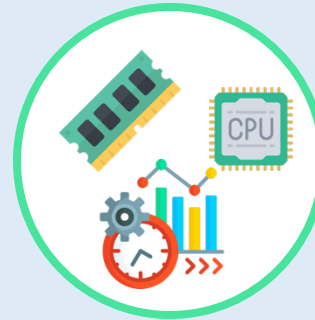
Frontend Node



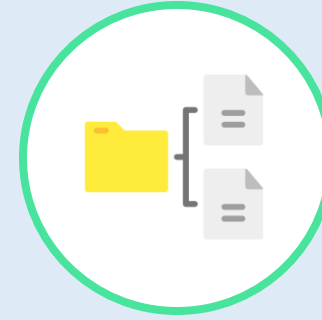
Function and Software



Remote
connection



Manage
resources



Shared storage



Manage
environments



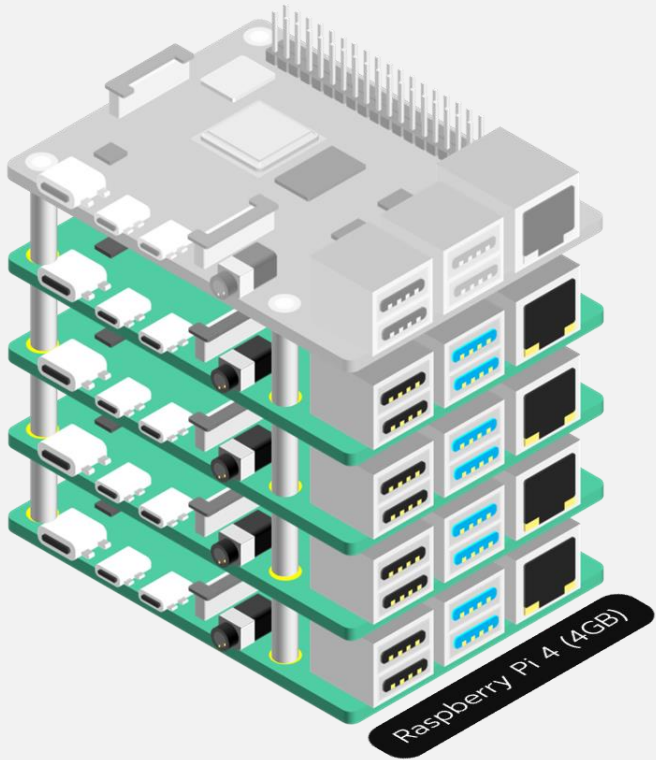
(Slurmctld, Slurmdbd)



NFS

nfs-kernel-server

Compute Node



Function and Software



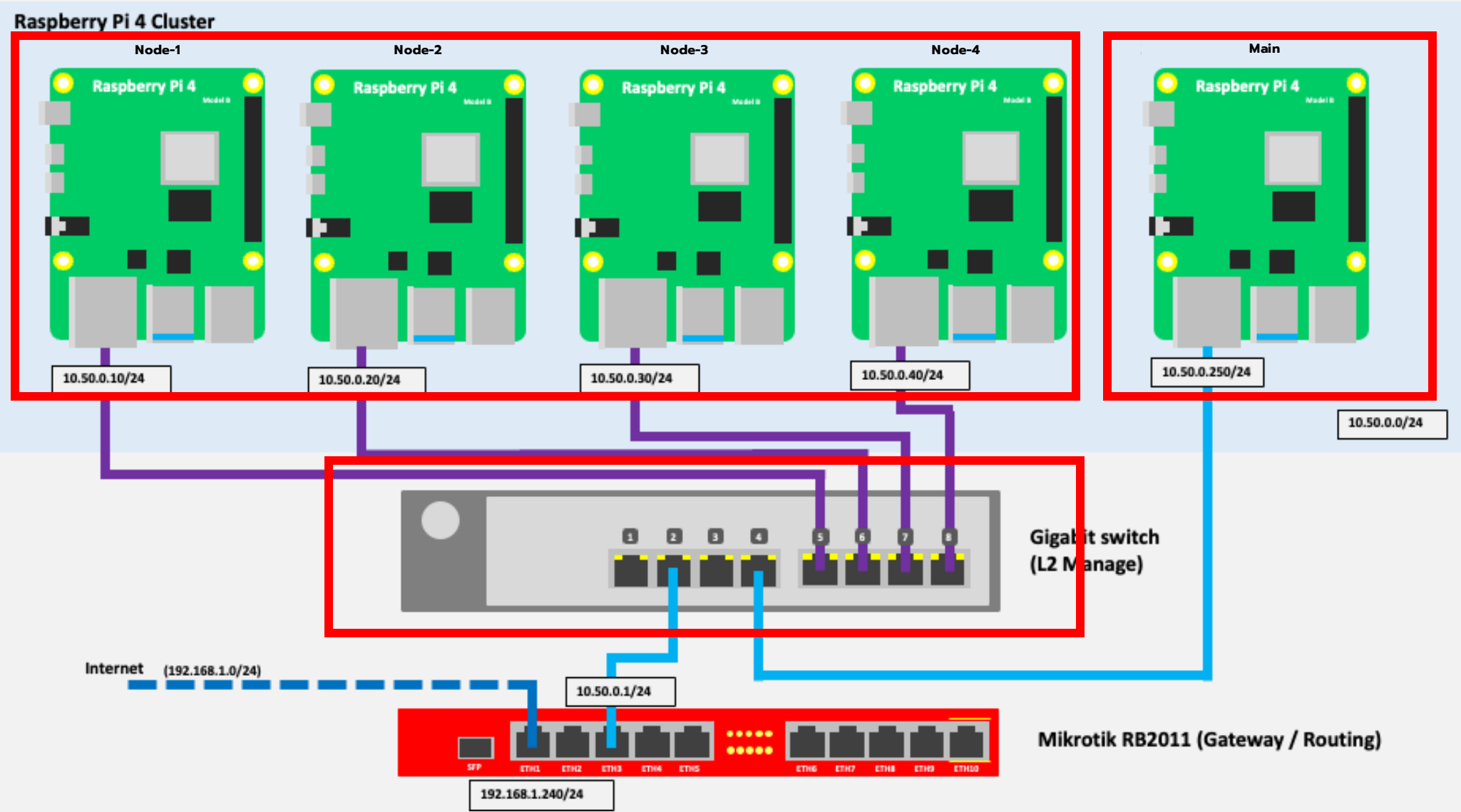
Task executing

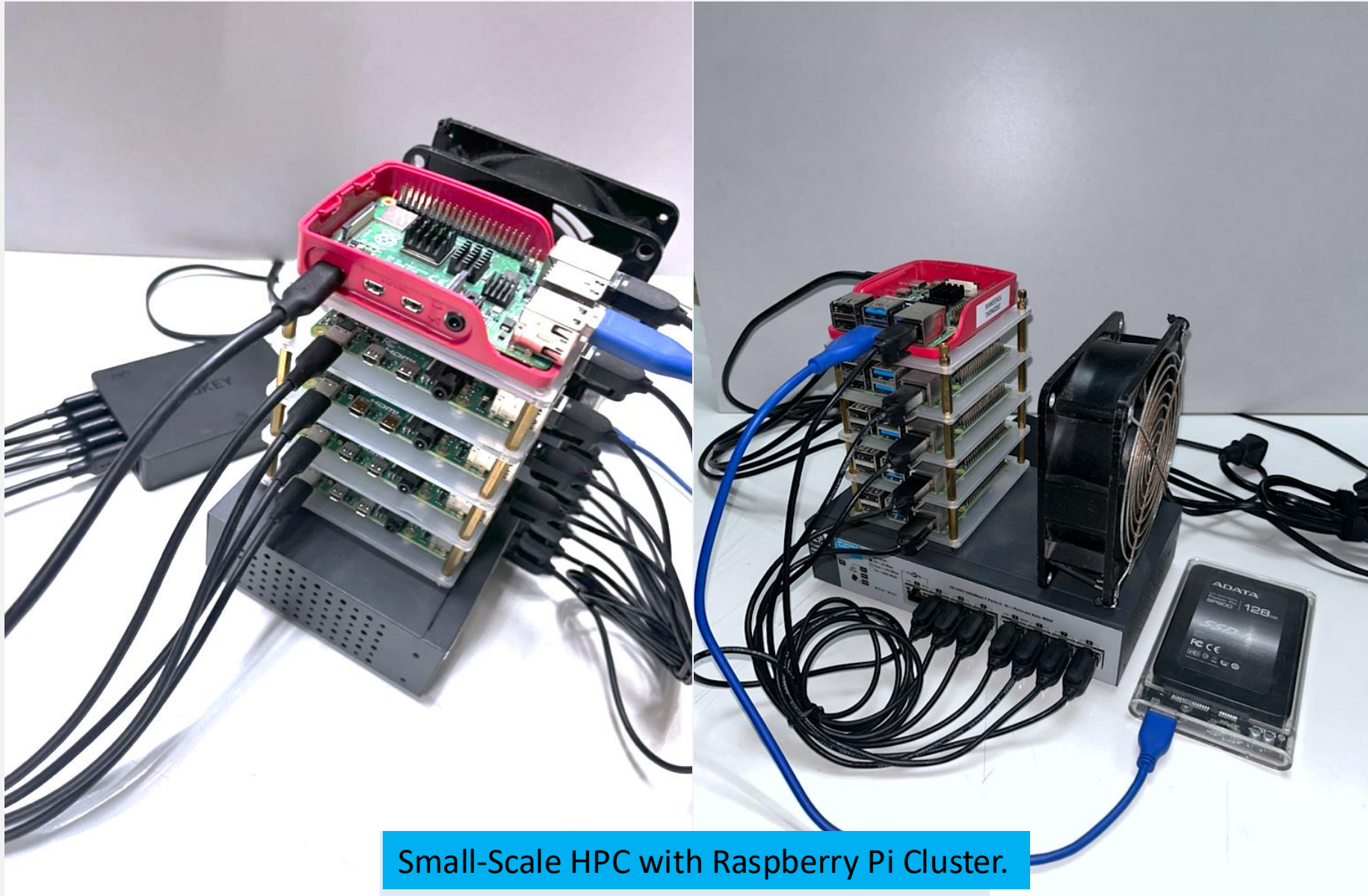


Slurmd

NFS

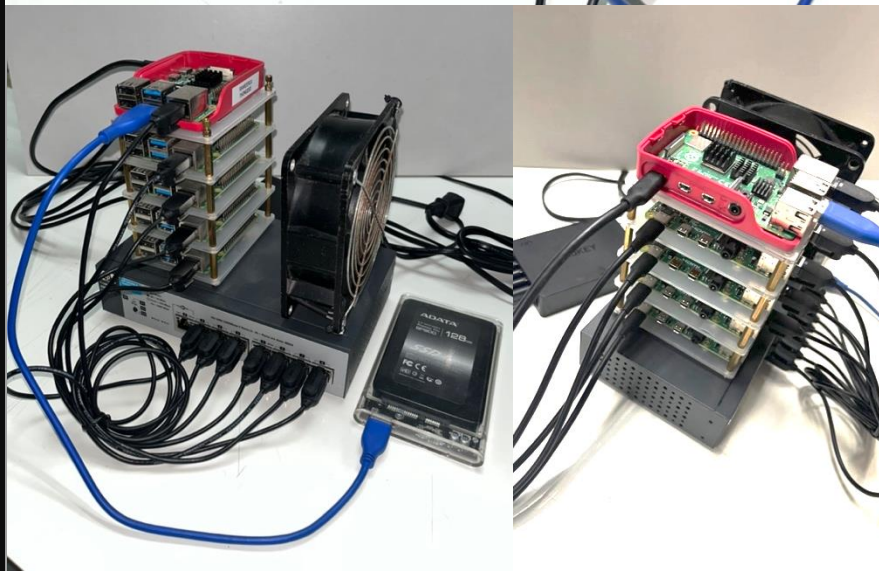
nfs-common





Small-Scale HPC with Raspberry Pi Cluster.

Evaluation



We evaluate the HPC using these applications

HPL

Numerical Computation

GROMACS

Molecular Dynamics (MD)
Simulation

MNIST

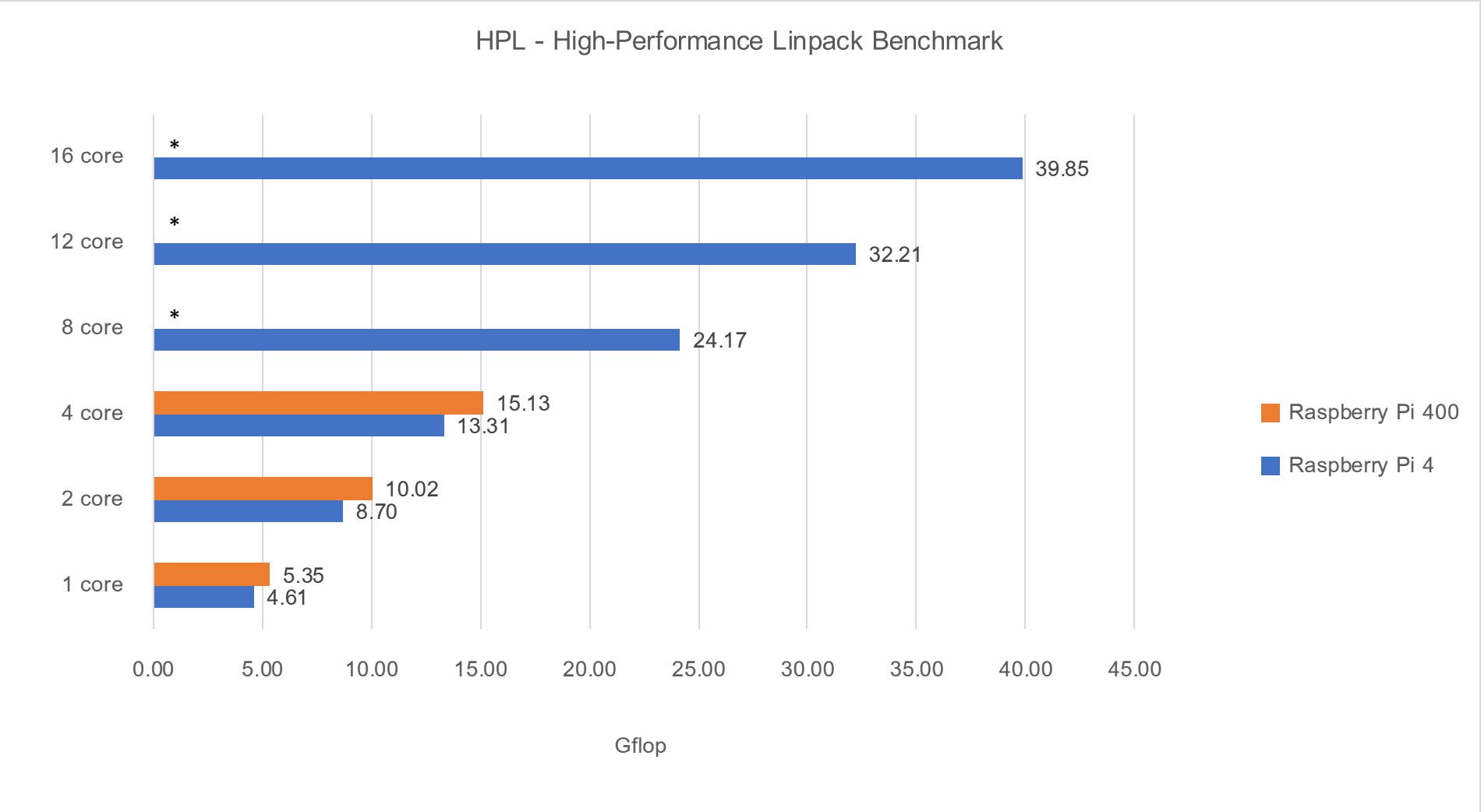
Artificial Intelligence (AI)
Training

System specification for evaluation :

Raspberry Pi 4 cluster	
CPU	Broadcom BCM2711 Cortex-A72 (ARM v8)
Clock Speed	1.50 GHz
Core	4
RAM	LPDDR4 - 4 GB
Connectivity	Wire Ethernet

Raspberry Pi 400 cluster	
CPU	Broadcom BCM2711 Cortex-A72 (ARM v8)
Clock Speed	1.80 GHz
Core	4
RAM	LPDDR4 - 4 GB
Connectivity	Wireless Ethernet

TARA cluster (Actual HPC)	
CPU	Intel(R) Xeon(R) Gold 6148 CPU
Clock Speed	2.40 GHz
Core	20
RAM	DDR4 - 192 GB
Connectivity	Infiniband



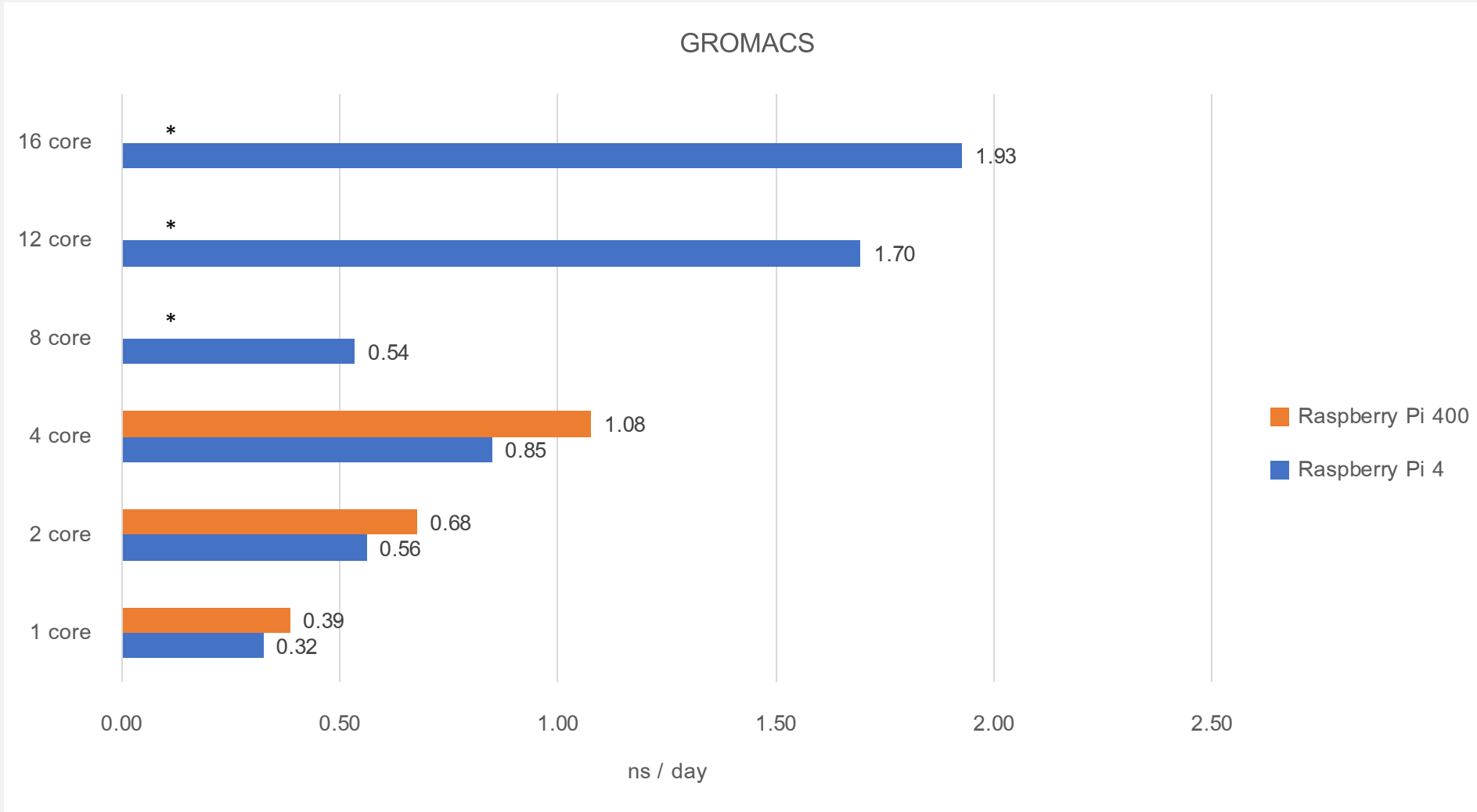
Configuration
GCC : 9.4.0
OpenMPI : 4.1.3
OpenBLAS : 0.3.2
HPL : 2.3
N : 20352



Higher is better

* Interconnection lantency problem

Evaluation : GROMACS



Configuration

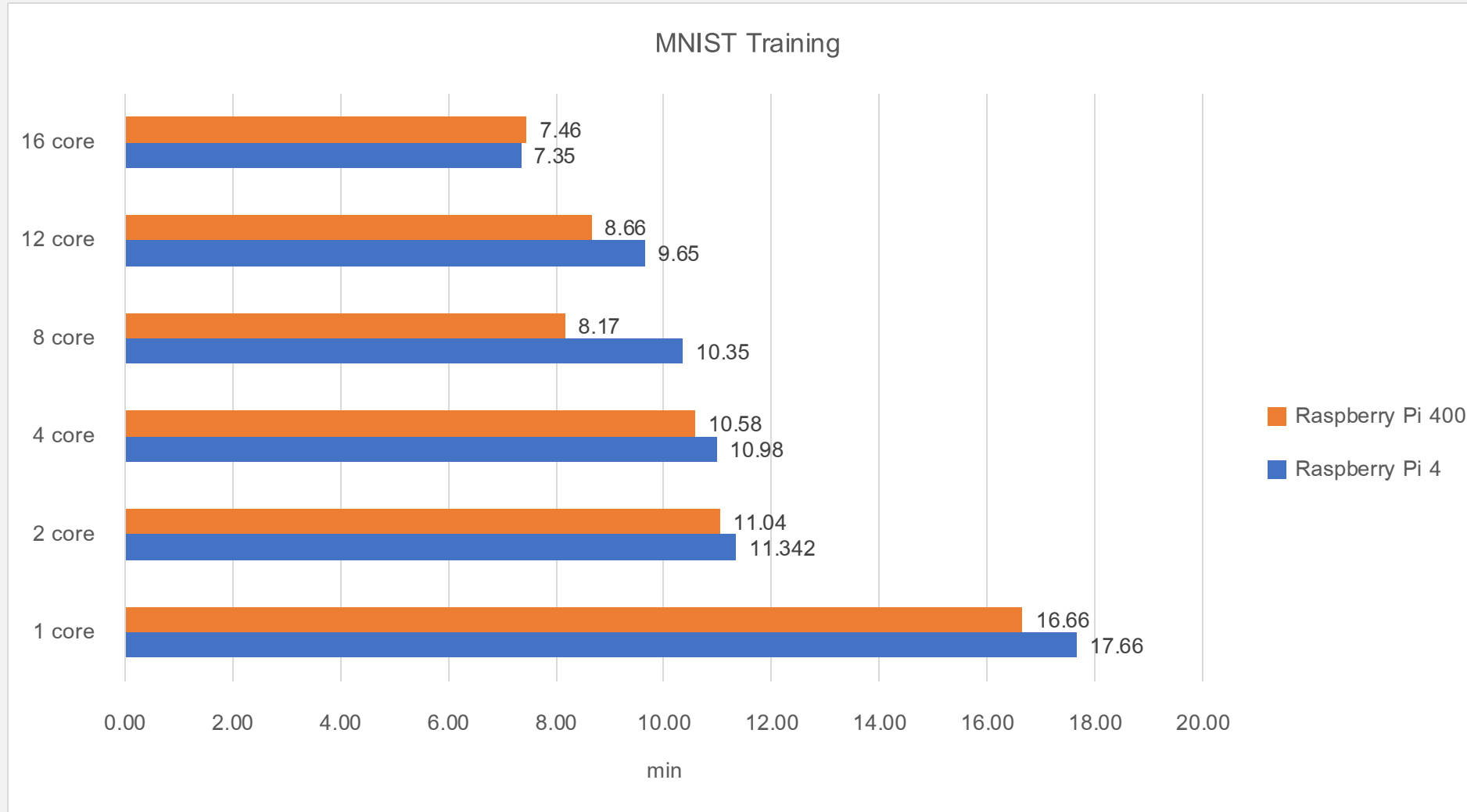
GCC : 9.4.0
OpenMPI : 3.1.6
OpenBLAS : 0.3.2
GROMACS : 2022.1
Dataset : HFA2A (36k atom)



Higher is better

* Interconnection latency problem

Evaluation : MNIST



Configuration

GCC : 9.4.0

OpenMPI : 3.1.6

OpenBLAS : 0.3.2

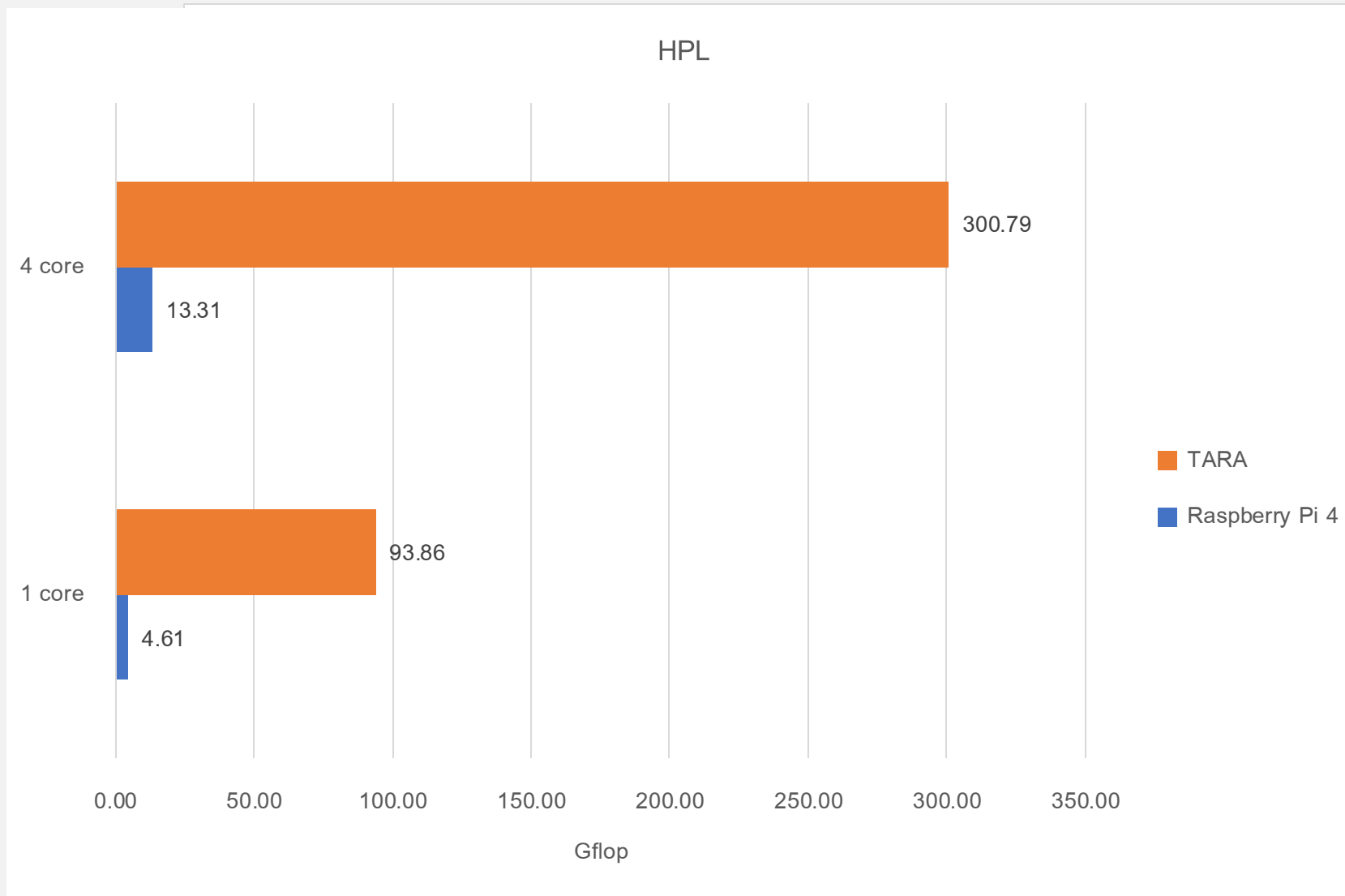
Horovod : 0.24.3

Dataset : MNIST (60000 image)



Lower is better

Evaluation : Single node performance



Configuration – Raspberry Pi 4

GCC : 9.4.0
OpenMPI : 4.1.3
OpenBLAS : 0.3.2
HPL : 2.3
N : 20352

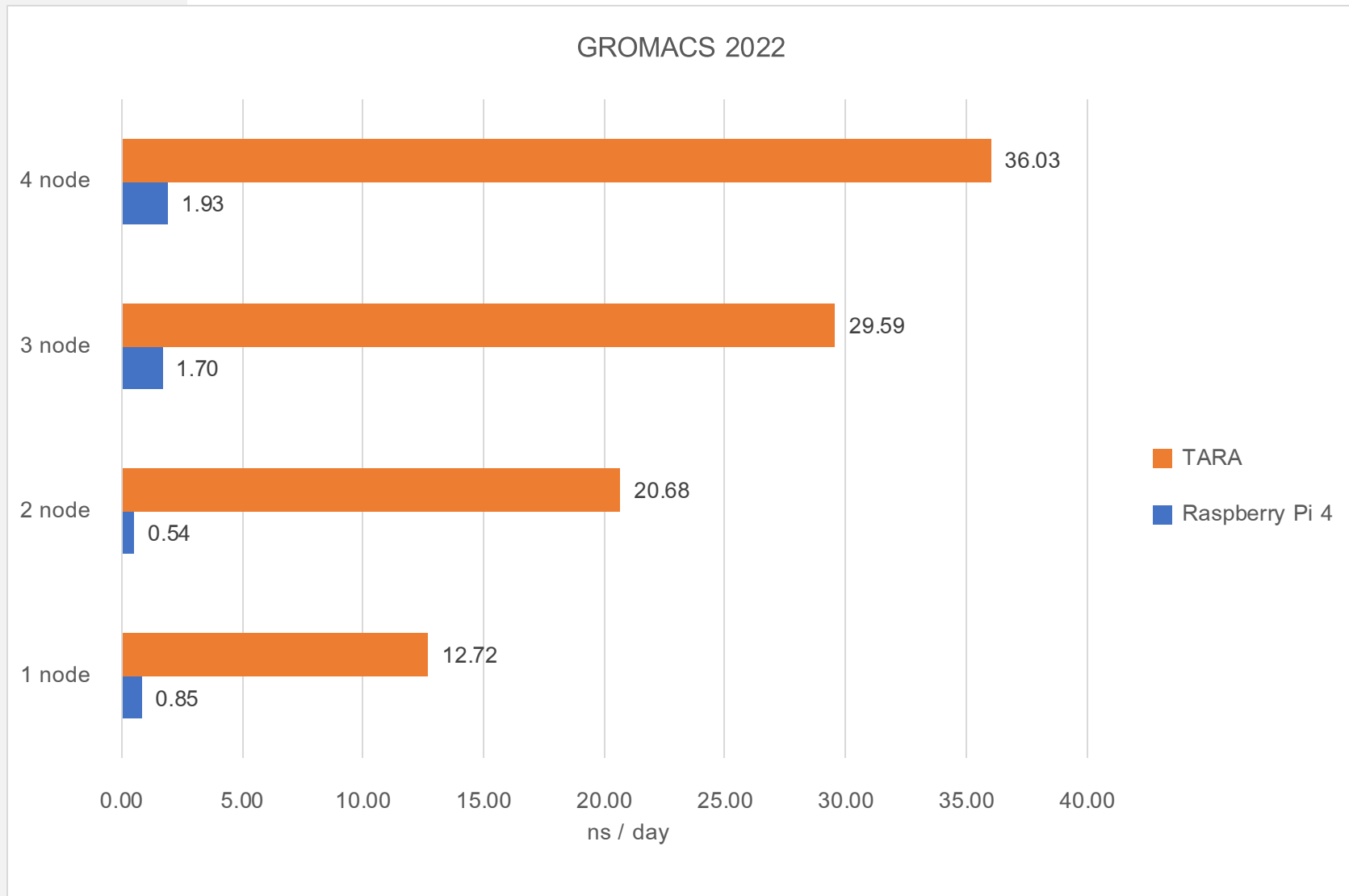
Configuration - TARA

GCC : 10.3.0
OpenMPI : 4.1.1
OpenBLAS : 0.3.15
HPL : 2.3
N : 20352



Higher is better

Evaluation : Multi-node performance



Configuration – Raspberry Pi 4

GCC : 9.4.0
OpenMPI : 3.1.6
OpenBLAS : 0.3.2
GROMACS : 2022.1
Dataset : HFA2A (36k atom)

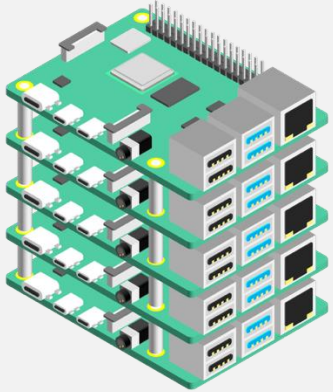
Configuration - TARA

GCC : 10.3.0
OpenMPI : 4.1.1
OpenBLAS : 0.3.2
GROMACS : 2022.1
Dataset : HFA2A (36k atom)



Higher is better

Conclusion



- ✓ Setup Small-Scale HPC with Raspberry Pi cluster complete
- ✓ Small-Scale HPC can run actual HPC application
- ✓ Small-Scale HPC can represent actual HPC



```
ubuntu@main:~$ sinfo
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
general*   up 7-00:00:00    4   idle node-[1-4]
high-mem   up 7-00:00:00    1   idle node-5
ubuntu@main:~$ squeue
JOBID PARTITION     NAME     USER ST
ubuntu@main:~$ |
```

The future work



The material and course for study



Performance tuning



Training for producing personnel and engineers

Thank you for the Knowledge, techniques, and methods that support this research. It is supervised by Dr.Putt Sakdhnagool and Dr.Viwan Jarerattanachat from the NSTDA Supercomputer center (ThaiSC).

Thanks to the JSTP-SCB scholarship program for internships in the National Science and Technology Development Agency (NSTDA).

And lastly, I would like to thank the Department of Computer Engineering, King Mongkut's Institute of Technology Ladkrabang (KMITL), for supporting research sites and resources.



Q & A

